

ABDESSAMAD ELIAS

SENIOR SOFTWARE DEVELOPMENT ENGINEER

PROGRAMING LANGUAGES

C • C++ • Java • C# • Go • Rust

SCRIPTING LANGUAGES

Javascript / Typescript • Python

CLOUD ENVIRONMENTS

AWS • Kubernetes

PLATFORMS AND FRAMEWORKS

Java Enterprise (Jakarta) • Spring •
Android Open Source Project

RELIABILITY AND OBSERVABILITY

CloudWatch • Prometheus •
OpenTelemetry

DATABASE SYSTEMS

PostgreSQL • AWS DynamoDB •
Cassandra • ElasticSearch

DATA STREAMING PLATFORMS

Kafka • AWS Kinesis • AWS SQS / SNS

DEV OPS

Terraform • Ansible

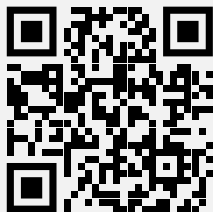
SECURITY

Threat Modeling • Authentication and
Authorization Workflows • Encryption and Secret
Management

LANGUAGES

English • French

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PROFESSIONAL SUMMARY

Senior Software Development Engineer with close to 12 years of experience building scalable distributed systems and cloud infrastructure at Amazon Web Services, Meta, and Murex. Proven track record of optimizing data replication services handling hundreds of petabytes across close to 40 regions, mentoring engineering teams, and architecting middleware solutions for high-frequency trading platforms. Expert in cloud-native architectures, performance optimization, and collaborating with cross-functional teams to deliver mission-critical systems.

PROFESSIONAL EXPERIENCE

■ SOFTWARE DEVELOPMENT ENGINEER

Amazon Web Services LLC (S3 Replication) • Toronto, ON - Canada • July 2022 - November 2024

Responsible for designing and implementing scalable solutions to enhance data durability and availability for AWS S3 replication services.

Optimized replication latency and traffic management across distributed systems, reducing operational load and improving incident response time by implementing advanced monitoring and automated remediation

Led technical mentorship program for junior and mid-level engineers, successfully onboarding them to complex distributed systems and S3 architecture.

Conducted comprehensive security audits and remediation plans across multiple S3 Replication services, identifying and resolving critical vulnerabilities while maintaining service availability.

Delivered cost optimization initiatives through intelligent traffic routing and resource management, reducing operational expenses while maintaining performance SLAs

■ SOFTWARE DEVELOPMENT ENGINEER

Meta Inc (Reality Labs) • Toronto, ON - Canada • December 2021 - July 2022

Designed and implemented advanced virtual camera system integrating rendering pipeline with Android Open Source Project (AOSP) hardware layer for next-generation AR/VR devices

Built AR/VR hardware integration solutions leveraging Android Open Source Project platform to create immersive experiences and optimize system performance for emerging Reality Labs products

Collaborated with cross-functional teams in fast-paced environment to develop scalable AR/VR platform architecture and enhance system performance metrics.

Contributed to technical design decisions for hardware-software integration, focusing on performance optimization and user experience enhancement

■ SOFTWARE DEVELOPMENT ENGINEER

Amazon Web Services LLC (Commerce Platform) • Vancouver, BC Canada • June 2019 - December 2021

Improved AWS Commerce Platform by developing robust, scalable software components supporting global transaction processing infrastructure

Designed and implemented critical features to improve transaction processing efficiency, reliability, and scalability for AWS billing and commerce systems

Collaborated with product managers and engineering teams to align development initiatives with business objectives while maintaining high code quality standards

Optimized system performance through architectural improvements and code optimizations, ensuring platform scalability for growing AWS customer base.

■ SOFTWARE DEVELOPMENT ENGINEER

Murex SAS • Paris, France • November 2013 – May 2019

Delivered foundational components of the Murex middleware framework integrating Murex financial platform with external market data providers and trading platforms, enabling financial data processing.

Improved system velocity and incident response capabilities by decoupling previously integrated Murex platform modules and enabling dedicated platform outsourcing

Served as technical advisor for distributed development team of external business logic developers, consultants, and integrators across multiple geographic locations.

Participated in design of scalable foundational services *framework* leveraging proprietary solutions and industry-standard technologies to support business integration workflows.

Participated in design and rollout of a service-oriented architecture (SOA) enabling modular deployments across on-premise datacenters and external cloud networks.

Worked on definition of service boundaries, contracts, and versioning strategy;

Introduction of API and schema governance to support iterative migration without breaking dependent consumers.

Collaborated on building a standardized service framework with cross-cutting concerns (identity/authorization, configuration, logging, retries, idempotency, circuit breakers) to accelerate new service delivery and ensure consistency across hybrid environments.

Implemented multiple mechanisms and integration points supporting enterprise integrations patterns.

Established observability foundations (distributed tracing, centralized logs, metrics, and SLOs) across on-premise and cloud services to support cross-boundary troubleshooting and reduce MTTR.

Partnered with external market data providers to adapt connectivity for hybrid topologies and workflows.

■ SOFTWARE DEVELOPMENT ENGINEER

IBM France Lab • Paris, France • February 2013 – September 2013

Developed components for WebSphere ILOG JRules BRMS (IBM Operational Decision Manager) to enhance decision-making processes within client organizations

Optimized business rule processing and defined performance indicator methodologies, enabling correlation inference across large business data volumes

Contributed to enterprise decision management solutions by implementing rule optimization algorithms and performance monitoring tools

ACADEMIC QUALIFICATIONS

■ MASTER OF ENGINEERING IN COMPUTER SCIENCE

Institut Polytechnique de Bordeaux France • September 2009 – September 2013

■ SCIENTIFIC PREPARATORY CYCLE TO FRANCE SCHOOLS OF ENGINEERING

Lycée Descartes Tours France • September 2007 – September 2009